DM UNIT-1 - Easy

Disaster Management (Jawaharlal Nehru Technological University, Kakinada)

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UNIT 1

UNDERSTANDING NATURAL DISASTERS

1.0 INTRODUCTION

Disasters can arise from both natural and human causes. They cannot be predicted, and once they occur, they need to be dealt with in a mature, tactful and responsible manner. A lot of on-the-spot decisions need to be taken and relief activities need to be organized and coordinated. In this unit, you will learn about various disasters and their impact on the living beings and the environment.

1.2 DISASTER AND ITS TYPES

A disaster is a mishap or hazard which causes huge loss of life and property and disrupts the balance of the economy. It is a tragic event with drastic consequences for living beings as well as social and individual development. A disaster can be caused by either natural or man-made factors. Both these factors need to be taken care of to prevent a disaster or lessen its impact. Disasters also arise due to inefficient management of risks. If a safety net is devised to address the potential risks, it would lead to reduction in damages triggered by disasters. Developing countries are more vulnerable to disasters.

An environmental disaster is a mishap or hazardous event which directly influences the environment, bringing serious alterations in the same. These alterations become the root cause of failures or damages that would occur following a disaster. Since the environmental disasters have direct impact on the environment, they could stunt economic growth, lead to socio-economic failures, deteriorate environmental conditions or threaten life. Apart from heavy immediate effects or damages, the environment also suffers from the long-term consequences of a disaster, which can prove to be costly. As these long-term effects alter the ecosystem, they can lead to more deaths over the next few years by giving rise to certain diseases and ailments.

They may also hinder tree growth or stop cultivation in a particular area, almost instantly. When a disaster strikes, the economy needs to divert all its resources towards the affected areas to try and save its elements from damage to the maximum extent possible. However, it takes high costs to recover the elements which been lost in or damaged by the disaster.

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Causes of Environmental Disasters

The common causes of environmental disasters are as follows—haphazard population growth, poor planning and environmental deterioration.

- **Haphazard population growth**: Population explosion is a disaster in itself. To enable the growing population lead a healthy and satisfied life, it is necessary to fulfil its essential needs such as food, housing, employment, educational and health facilities. However, governments, particularly in developing countries, find it difficult to fulfil these needs of all their citizens. The non-fulfilment of essential human needs makes developing countries more vulnerable to natural disasters.
- **Poor planning**: Planners and developers have not been able to plan effectively to take care of the essential needs of the growing population. It is for this reason that more and more people have been left unprotected and vulnerable to disasters.
- **Environmental deterioration**: Gradual deterioration of the environment and the tools that nature has provided us as protection against disasters has also emerged as a major cause of disasters. If natural remedies for and protections against disasters are removed, then man-made protections need to be provided. The intensity of a disaster in an area is inversely proportional to the natural and man-made protection the area has, and thus it becomes necessary to ensure that such a protection remains intact.

1.2.1 Types of Disasters

Broadly, disasters have been categorized as natural and man-made disasters on the basis of their causes. The cause-based disaster classification facilitates altering or stopping altogether such activities that lead to a disaster. Taking such steps would help in reducing the impact of a disaster even if it strikes. The magnitude of a disaster also decides the kind of damage it would cause and the steps required to avoid or lessen the damage. Therefore, disasters are also classified on the basis of their magnitude. Disasters which cause large-scale loss are called major disasters while those which do not cause heavy devastation are called minor disasters.

Natural disasters: Natural disasters are those hazardous events which occur naturally. They are caused by sudden changes in the environment. Natural disasters include earthquakes, cyclones, volcanic eruptions, forest fires, tornadoes. They can cause heavy loss of life and damage to physical structures, leading to huge financial losses. The magnitude of these

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disasters may vary. Every area is susceptible to its own set of natural disasters and thus it is important to take preventive measures accordingly. For example, the areas where tectonic plates are too close would be susceptible to earthquakes, whereas areas near volcanic formations would be more likely to suffer from volcanic eruptions. Governments must make necessary arrangements to protect people from area-specific natural disasters. This would ensure minimum damage. One effective way to reduce the impact of disasters is to use latest technology. For example, infrastructure and buildings can be strengthened to withstand natural disasters by using new construction technology and materials. Natural disasters are also known as geographical disasters, which are discussed ahead in detail.Natural disasters can be of various kinds, the most prominent of which, have been listed as follows:

Land-movement disasters: These kinds of disasters can further be classified into the following:

- Avalanches
- Earthquakes
- Landslides and mudflows
- Volcanic eruptions

Water disasters: The various water disasters are:

- Floods
- Limnic eruptions
- Tsunamis

Weather disasters: The disasters caused by weather disturbances are:

- Blizzards
- Cyclonic storms
- Droughts
- Hailstorms
- Tornadoes

Natural fires (like forest fire)

Health and diseases: The hazards that can be caused due to health problems are as follows:

- Epidemic

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- Famineods

Man-made disasters:

As is clear from their name, man-made disasters are caused by human intervention or activities. These may be dangerous to life, physical elements or economic components of the environment and the resultant damage could prove disastrous for the whole economy. Man-made disasters occur due to a variety of reasons. One reason is the hardened human attitudes and approaches to view things and situations in certain ways. These lead to man-made disasters such as big crimes, arson, civil disruption, war, and terrorism. Another type of man-made disasters includes those hazardous events which are caused by technological faults or breakdowns. These disasters include industrial fires, structural collapse, chemical or gaseous release and accidents involving transport means such as cars, planes, ships, trainsor space shuttles. Better technology, sufficient precautions and careful working with technology are the only steps which can prevent or lessen the damage from technology-related disasters. With the growing climatic changes and unstable landforms all over the world, human beings are becoming more vulnerable to disasters and hazards. The drastic changes in the weather patterns have also led to a number of occurrences of the disasters. Technological advancements and the growing population density have also contributed to the world becoming increasingly unsafe.

Anthropogenic disasters:

Anthropogenic disasters are threats that have an element of human intent, negligence or error or have witnessed failure of a man-made system. They are also known as man-made disasters since they are the result of a failing or error on the part of humans.

Anthropogenic disasters can be classified into the following categories:

Sociological hazards: The disasters that are caused due to sociological factors are:

- Crime
- Arson
- Civil disorder
- Terrorism
- War

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Technological hazards: Technological advancements can lead to the following disasters:

- Industrial hazards
- Structural collapse
- Power outage
- Fire

Hazardous materials: The chemical substances that can cause disasters are as follows:

- Radiation contamination
- CBRNs

1.3 FLOODS:

Floods are caused by too much rain or water in a location, when the excess water surpasses the limits of its confines. Generally, occurrences of floods are more frequent in low-lying lands. Some plains may periodically flood; known as a flood plain. Ancient Egypt relied on these floods for agriculture. Floods present a significant danger with enough force to sweep away massive objects such as houses, cars and trees.

Many reasons could lead to a flood, including prolonged rainfall from a storm, thunderstorms, rapid melting of snow, overflowing rivers from excess rain, bursting of man-made dams or levees. Monsoon rainfalls can also cause floods, such as in Bangladesh due to extended periods of rainfall. There is a growing feeling that the incidence and intensity of floods has grown alarmingly over the years. A major cause is the increased encroachment of flood plains because of development and population pressure.

The damage caused by floods can at best be minimized and not altogether eliminated or in other words, there can really be no such thing as ‘fool proof protection’ or ‘absolute flood control’ for all magnitudes of floods. The concept of flood management, therefore, aims for such planned measures which ensure profitable and economic utilization of the flood plains for the benefit of mankind and at the same time emphasizing that during high floods, there is no severe damage as far as possible. An essential part of a disaster preparedness plan is education for those who may be threatened by disaster.

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Types of Floods

1. Flash floods: If extremely heavy rainfall occurs for some time and the ground loses its absorbing capacity then this type of flood generally takes place. These types of floods last for a short duration with a relatively high peak discharge.

2. River floods: If due to heavy rain the water level of river flows much above the danger mark, this type of flood occurs.

3. Storm surge: Low-pressure storm associated with strong wind causes the sea level to rise Suddenly . If strong winds are directed towards the shore, they drive the seawater onto the land. The storm that produces the surge can also give rise to heavy rainfall, resulting in coastal areas being subjected simultaneously to flooding due to incessant rains and seawater.

Factors that cause floods

Some of the factors responsible for causing floods are listed below:

- Rivers in spate due to heavy and continuous rain for a long period such as few days or weeks.
- Flowing of water much above its danger level due to inadequate capacity within the banks of the river to contain high flows.
- Breaching of embankments
- Breaching of water reservoir
- Storm surges
- Snowmelt
- River bank erosion and silting of riverbeds
- Landslides leading to obstruction of flow and change in the river course.
- Poor natural drainage
- Flow retardation due to tidal and backwater effects.
- Deforestation
- Cyclone
- The resistance to the flow of water due to various reasons

Among all the natural disasters affecting India, frequent river floods are the most devastating, which cause maximum damages of life and property. Total flood prone areas in India are 40

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million hectares, out of which 9.4 per cent falls in Assam. Besides draught, about 90 per cent damages to crops are only due to flood.

1.4 DRAUGHTS

A drought is an extended period of months or years when a region notes a deficiency in its water supply. Although droughts can persist for several years, even a short, intense drought can cause significant damage and harm the local economy.

Causes

Generally, droughts occur when a region receives consistently below-average precipitation. It can have a substantial impact on the ecosystem and agriculture of the affected region. This global phenomenon has a widespread impact on agriculture. Generally, rainfall is related to the amount of water vapour in the atmosphere, combined with the upward forcing of the air mass containing that water vapour. If either of these are reduced, the result is a drought. This can be triggered by an above average prevalence of high pressure systems, winds carrying continental, rather than oceanic air masses (ie. reduced water content),and ridges of high pressure areas form with behaviours which prevent or restrict the developing of thunderstorm activity or rainfall over one certain region. Oceanic and atmospheric weather cycles such as the El Niño-Southern Oscillation (ENSO) make drought a regular recurring feature of the Americas along the Pacific Coast and Australia.

Human activity can directly trigger exacerbating factors of droughts such as over-farming, excessive irrigation, deforestation, and erosion adversely impacting the ability of the land to capture and hold water. While these tend to be relatively isolated in their scope, activities resulting in global climate change are expected to trigger droughts with substantial impact on agriculture throughout the world, especially in developing nations. Overall, global warming will result in increased world rainfall. Along with drought in some areas, flooding and erosion will increase in others. Paradoxically, some proposed solutions to global warming that focus on more active techniques, solar radiation management through the use of a space sunshade for one, may also carry with them increased chances of drought.

Consequences

Periods of drought can have significant environmental, agricultural, health-related economic and social consequences. The effect varies according to vulnerability. For example,

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subsistence farmers are more likely to migrate during drought because they do not have alternative food sources. Areas with population that depend on subsistence farming as a major food source are more vulnerable to drought-triggered famine. Drought is rarely, if ever, the sole cause of famine; socio-political factors such as extreme widespread poverty play a major role. Drought can also reduce water quality, because lower water flows reduced illusion of pollutants and increase contamination of remaining water sources. A few common consequences of drought include:

- Diminished crop growth or yield productions and carrying capacity for livestock
- Dust bowls, themselves a sign of erosion, further erode the landscape
- Dust storms occur when drought hits an area suffering from desertification and erosion
- Famine due to lack of water for irrigation
- Habitat damage, affecting both terrestrial and aquatic wildlife
- Malnutrition, dehydration and related diseases
- Mass migration, resulting in internal displacement and international refugees
- Reduced electricity production due to insufficient available coolant for power stations and reduced water flow through hydroelectric dams.
- Shortages of water for industrial users
- Social unrest
- War over natural resources, including water and food
- Wildfires, such as Australian bushfires, are more common during times of drought.

1.5 CYCLONES

A tropical cyclone is a storm system characterized by a large low-pressure centre and numerous thunderstorms that produce strong winds and heavy rain. Tropical cyclones feed on heat released when moist air rises, resulting in condensation of water vapour contained in the moist air. The term ‘tropical’ refers to both the geographic origin of these systems, which form almost exclusively in tropical regions of the globe, and their formation in maritime tropical air masses. The term ‘cyclone’ refers to such storms’ which are cyclonic in nature, with counter clockwise rotation in the Northern Hemisphere and clockwise rotation in the Southern Hemisphere. Depending on its location and strength, a tropical cyclone is referred to by names such as hurricane, typhoon, tropical storm, cyclonic storm, tropical depression, and simply cyclone. In the Atlantic and northern Pacific, the storms are called ‘hurricanes,’

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after the Caribbean god of evil, named Hurrican. In the north western Pacific, the same powerful storms are called 'typhoons.'

In the south eastern Indian Ocean and south western Pacific, they are called 'severe tropical cyclones'. In the northern Indian Ocean, they're called 'severe cyclonic storms.' In the south western Indian Ocean, they're just 'tropical cyclones.'

Why do cyclones occur?

- When warm air rises from the earth and condenses to form clouds, a great amount of heat is released. The combination of this heat and moisture often leads to thunderstorms, from which a tropical storm can develop.
- The trigger for most Atlantic hurricanes is an easterly wave, a band of low pressure moving westwards, which may have begun as an African thunderstorm.
- Typhoons in the Far East and Cyclones in the Indian Ocean often develop from a thunderstorm in the equatorial trough. During the hurricane season, the Coriolis Effect of the Earth's rotation leads to the winds in the thunderstorm spinning in a circular motion.

Effects

While tropical cyclones can produce extremely powerful winds and torrential rain, they are also able to produce high waves and damaging storm surge as well as spawning tornadoes. They develop over large bodies of warm water, and lose their strength if they move over land. This is why coastal regions can receive significant damage from a tropical cyclone, while inland regions are relatively safe. Heavy rains, however, can produce significant flooding inland, and storm surges can produce extensive coastal flooding up to 40 kilometres from the coastline. Although their effects on human population can be devastating, tropical cyclones can also relieve drought conditions. They also carry heat and energy away from the tropics and transport it toward temperate latitudes, which make them an important part of the global atmospheric circulation mechanism. As a result, tropical cyclones help to maintain equilibrium in the Earth's troposphere, and a relatively stable and warm temperature worldwide.

Tropical cyclones out at sea cause large waves, heavy rain, and high winds, disrupting international shipping and, at times, causing shipwrecks. Tropical cyclones stir up water,

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leaving the air cooler behind them, which causes the region to be less favourable for subsequent tropical cyclones. On land, strong winds can damage or destroy vehicles, buildings, bridges, and other outside objects, turning loose debris into deadly flying objects. The storm surge, or the increase in sea level due to the cyclone, is typically the worst effect from tropical cyclones, historically resulting in 90 per cent of tropical cyclone deaths. The broad rotation of a tropical cyclone, and vertical wind shear at its periphery, spawns tornadoes. Over the past two centuries, tropical cyclones have been responsible for the deaths of about 1.9 million people worldwide. Large areas of standing water caused by flooding lead to infection, as well as contributing to mosquito-borne illnesses. Crowded evacuees in shelters increase the risk of disease. Tropical cyclones significantly interrupt infrastructure, leading to power outages, bridge destruction, and the hampering of reconstruction efforts.

1.6 NATURAL /GEOGRAPHICAL DISASTERS

Natural/geographical disasters have already been discussed in the earlier section. In this section, we will discuss the different types of geographical disasters.

1.6.1 Earthquakes

Earthquakes are caused by a sudden shift or movement deep underground in the Earth's tectonic plates, causing the Earth's crust to shake violently, with vibrations varying in magnitude. On the surface, we see this as the shaking of the ground, causing damage to poorly built structures. Earthquakes occur unpredictably along fault lines and are capable of killing thousands of people. The most powerful earthquakes can destroy even the best built structures. Earthquakes can also cause other disasters including tsunamis and volcanic eruptions. Seismometers can detect the strength of an earthquake. In the past, seismologists used to estimate earthquake intensity using the Richter scale devised by Charles Richter. However, today, the moment magnitude scale, which is an improved version of the Richter scale, is used by seismologists to measure the size of earthquakes in terms of the energy released. An earthquake has point of origin underground called 'focus'. The point directly above the focus on the surface is called the 'epicentre'. Earthquakes by themselves rarely kill people or wildlife. It is usually the secondary events that they trigger, such as building collapse, fires, tsunami (seismic sea waves) and volcano. Many of the disasters related to an earthquake are actually human disasters and could be avoided by better construction, safety

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systems, early warning and evacuation planning, the term unnatural disaster is not unwarranted.

Some of the most recent and disastrous earthquakes in recent times are:

- The 2011 Japanese Earthquake, registering a magnitude of 9.0, triggered a massive tsunami. Over 15,000 people were killed in the disaster.
- The 2010 Haiti Earthquake, registering a magnitude of 7.0, killed an estimated 100,000-150,000 people.
- The 2004 Indian Ocean earthquake, the third largest earthquake in recorded history, registered a moment magnitude of 9.3. The huge tsunami triggered by the earthquake cost the lives of at least 229,000 people in 14 countries.
- The 2001 Gujarat Earthquake, measuring 7.7 on the moment magnitude scale, killed at least 20,000 people.

1.6.2 Landslides

Landslides are an extremely frequent geological event. They occur when masses of rock, earth, or debris move down a slope, caused by disturbances in the natural stability of a slope. They are defined as downslide of soil and rock, which happen due to natural phenomena or man-made actions. There are various kinds of movements like falls, slides, topples, lateral spreads, and flows. Landslides generally occur in hilly terrains. In India, they usually occur in the Himalayan region as the Himalayas are geologically young and susceptible to earthquakes and intensive soil erosion. Landslides also occur in the Western Ghats, Eastern Ghats and the Nilgiri hills with lesser frequency and intensity. Over the years, due to increasing human activity, the incidences of landslides have shown a disturbing upward trend of occurrence with higher damage to life and property.

Landslides take place when there are disturbances in the natural stability of a slope. They can accompany heavy rains or follow droughts, earthquakes, or volcanic eruptions. Mudslides develop when water rapidly accumulates in the ground and results in a surge of water-saturated rock, earth, and debris. Mudslides usually start on steep slopes and can be activated by natural disasters. Areas where wildfires or human modification of the land have destroyed vegetation on slopes are particularly vulnerable to landslides during and after heavy rains. Landslides are often secondary effects of heavy storms, volcanic eruptions and earthquakes. They cause high mortality, killing many thousands by burying villages and hillside houses,

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by sweeping vehicles off the road into ravines. Death mostly results from trauma and suffocation by entrapment under debris. Significant damage to property is also caused like a breakdown in the water systems, constructions, and lines of communication, and damaged crops.

Factors that cause landslides are often man-made, like intense deforestation, soil erosion, construction of human settlement in landslide prone areas, roads or communication lines in mountain areas, buried pipelines, among others.

Apart from landslides there are also:

- Mudslides: Debris flows, also known as mudslides, are a common type of fast-moving Landslide that tends to flow in channels.
- Lahars: A lahar is a volcanic mudflow or landslide. The 1953 Tangiwai disaster was caused by a lahar, as was the 1985 Armero tragedy, in which the town of Armero was buried and an estimated 23,000 people were killed.

1.6.3 Avalanches

Avalanche can be defined as a swift flow of snow down a slope, which may have been triggered naturally or due to human activity. An avalanche typically happens in mountainous regions, and also brings down air and water along with snow. A strong enough avalanche can drag down with itself ice, and large objects like rocks, trees, and other material down the slope. While rock slides, mudslides, serac collapses and rock avalanches are all caused by icefalls, an avalanche is different from them because it is primarily composed of flowing snow. In the mountains, avalanches are one of the commonest threats faced by life and property, mainly because they are rapid and have enough power to carry down other objects with them.

Usually, avalanches are categorized according to their size and other shape-related characteristics and rated according to the destruction they are capable of causing. Some traits of an avalanche used to categorize it include nature of the failure, sliding surface, type of snow involved, propagation mechanism of the failure, reason for trigger of avalanche, direction, slope elevation and slope angle. Avalanche size, mass, and destructive potential are

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rated on logarithmic magnitude scales, which usually consist of 4 to 7 categories, which vary with the system of observation being used and/or the region in which forecast is being done.

1.6.4 Volcanic Eruptions

An opening in the surface of the earth that allows lava, volcanic ash and gases to escape from its magma chamber below the surface is called a volcano. The sudden occurrence of a violent discharge of steam and volcanic material from a volcano is a volcanic eruption. There are essentially three meta types of volcanic eruptions. The most common are magmatic eruptions, which involve the decompression of gas within magma that propels it forward. Another type of volcanic eruptions are known as phreatomagmatic eruptions. These are driven by the compression of gas within magma, direct opposite of the process powering magmatic activity. The third type of volcanic eruption is known as phreatic eruption. It occurs as a result of the superheating of steam via contact with magma. Phreatic eruptions often exhibit no magmatic release, instead causing the granulation of existing rocks.

There are two types of volcanic eruptions in terms of activity—explosive eruptions and effusive eruptions. Explosive eruptions are characterized by gas-driven explosions that propel magma and tephra. Effusive eruptions, meanwhile, are characterized by outpouring of lava without significant explosive eruption. The most dangerous eruptions are explosive eruptions; in the history of the earth some explosive eruptions are thought to have caused climate changing events such as the Lake Toba eruption in Indonesia that occurred 69,000 years ago.

Hazards

Volcanic eruptions are extremely dangerous. Hazards during volcanic eruptions include emission of volcanic gases like carbon dioxide, sulfur dioxide, hydrogen sulfide, hydrogen, and fluorine. When sulfur dioxide gases react with water droplets in the atmosphere, it creates acid rain that corrodes vegetation. Carbon dioxide concentrations are poisonous to people and animals alike. Fluorine is a toxic gas that is absorbed onto volcanic ash which falls to the ground during eruptions. These ashes can poison livestock as well as water supplies. Along with volcanic gases, lava flows are another hazard of volcanic eruptions. Lava flows are molten rocks that ooze onto the earth’s surface. They are hot enough to burn everything in its path. Pyroclastic flows, or high-speed avalanches of hot ash, rock fragments, and gas, are other hazards of volcanic eruptions. Pyroclastic flows are extremely fast moving and have

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temperatures that exceed 1000°C. They can cause apocalyptic devastation during explosive eruptions.

For example, the eruption of Mount St. Helens on May 18, 1980, caused a huge pyroclastic surge which completely devastated an area of 230 square miles. Other dangers during volcanic eruptions are earthquakes, landslides and mudslides. When Pyroclastic flows mix with water, they can cause mudflows, also known as lahars.

Some of the most significant volcanic eruptions of recent times include:

- On May 18th, 1980, Mount St. Helens erupted in the United States causing 57 deaths and over a billion dollars in damage.
- On November 13th, 1985, an eruption on Nevado del Ruiz produced an enormous lahar that buried and destroyed the town of Armero in Tolima, killing around 25,000 people.
- On 10th April 1815, Mount Tambora on the island of Sumbawa in Indonesia erupted. The eruption was the largest in the recorded history of man. An estimated 90,000 people were killed as a direct result of the explosion or due to the aftermath. The eruption caused global Anomalies in the climate. As a result of the eruption, the year 1816 came to be known as the 'Year Without a Summer'.

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